

Trace-Tempered Choice

A Lean-faithful proof of Theorem 1

Daniele Condorelli

Standing notation. All payoff, action, and record alphabets are finite, and action and record alphabets are nonempty. The primitive relations, common-payoff block environments, and Axioms (A1)–(A8) are those in the paper. A payoff-preserving record processor T and an action report S satisfy, respectively,

$$(KT)(o, r' \mid a) = \sum_r K(o, r \mid a)T(r' \mid o, r), \quad (1)$$

$$(qS)(b)\widehat{K}(o, r \mid b) = \sum_a q(a)S(b \mid a)K(o, r \mid a). \quad (2)$$

The second identity is imposed without division and hence leaves rows at null reported actions unrestricted.

A Proof of Theorem 1

The only substantial result imported from elsewhere in the paper is the pure-record characterisation in Appendix D. We also use the ordinary mixture-space theorem: a complete and transitive mixture order that satisfies independence and admits calibration on every closed mixture segment has an affine representation. The proof identifies the value of changing one deterministic terminal payoff and then telescopes. An arbitrary joint channel is reduced to this case by sequentialisation.

A.1 Preliminaries

We first dispose of three pieces of bookkeeping. They will be used without further comment.

Recall that $(q, K) \succeq (p, L)$ denotes the paper’s canonical two-block comparison. For channels E, G at a common prior q , write $E \succeq_q G$ for $(q, E) \succeq (q, G)$. A *finite experiment at q* is a finite payoff–record channel paired with that prior; its marked terminal law is denoted by η_E .

Lemma 1 (Blocks, supports, and dummies). *Suppose Axioms (A1)–(A5) hold.*

- (a) *After identifying alternatives related by invertible action or record relabellings, the derived comparison of pairs is complete and transitive. If either member of a pair comparison is replaced by an indifferent alternative, the comparison is unchanged.*
- (b) *If $S = \text{supp } q$, then (q, K) is indifferent to the restriction of (q, K) to S .*
- (c) *If $s \in \Delta(B)$ and $K^\uparrow(o, r \mid a, b) := K(o, r \mid a)$, then*

$$(q \otimes s, K^\uparrow) \sim (q, K). \quad (3)$$

Proof. Place any finite collection of alternatives in one finite block environment. Axiom (A3) removes unused blocks, and Axiom (A1) supplies completeness and transitivity there. Invertible action and record relabellings are neutral by applying Axioms (A4) and (A5) in both directions. In particular, swapping both tags identifies the two orientations of a two-block comparison. This proves part (a); a three-block environment gives transitivity and a four-block environment gives replacement.

For part (b), report an action in S by its inclusion in A . In the other direction, report $a \in S$ when the realised action is a , and choose an arbitrary element of S off the support. Both reports

satisfy (2); rows attached to null reported actions may be completed arbitrarily. Axiom (A5) gives the two comparisons.

For part (c), first project (a, b) to a . Conversely, given a , draw b independently from s and report (a, b) . Again the two joint-law identities in (2) are exact, so Axiom (A5) applies in both directions. Part (a) completes both arguments. $\square$

Fix for the moment a full-support prior $q \in \Delta(A)$. For a joint channel K , write

$$p_K(o, r) := \sum_a q(a) K(o, r | a), \quad \pi_{or}^K(a) := \frac{q(a) K(o, r | a)}{p_K(o, r)}$$

when $p_K(o, r) > 0$, and define its *marked terminal law*

$$\eta_K := \sum_{(o, r): p_K(o, r) > 0} p_K(o, r) \delta_{(o, \pi_{or}^K)}. \quad (4)$$

It records the terminal payoff and the posterior about the realised action. Write $I_q(E)$ for the mutual information between the action and the complete visible output of E under q . When the output is a first-stage record Y , we also write this quantity as $I_q(A; Y)$.

Lemma 2 (Terminal laws and affine fibres). *Under Axioms (A1)–(A6), the following statements hold.*

- (a) *If q has full support and $\eta_K = \eta_L$, then $(q, K) \sim (q, L)$.*
- (b) *At every full-support q , the quotient of finite experiments by their marked terminal laws has an affine representation W_q .*

Proof. We give the finite Blackwell argument behind part (a). For a payoff o and posterior π , let

$$M_K(o, \pi) := \sum_{\substack{r: p_K(o, r) > 0 \\ \pi_{or}^K = \pi}} p_K(o, r).$$

Equality of marked laws gives $M_K(o, \pi) = M_L(o, \pi) =: M(o, \pi)$. For $p_K(o, r) > 0$, draw a record s of L according to

$$T(s | o, r) := \mathbf{1}_{\{p_L(o, s) > 0, \pi_{os}^L = \pi_{or}^K\}} \frac{p_L(o, s)}{M(o, \pi_{or}^K)}. \quad (5)$$

Choose an arbitrary probability row when $p_K(o, r) = 0$. This is a payoff-preserving record processor. Bayes' formula gives

$$q(a) K(o, r | a) = p_K(o, r) \pi_{or}^K(a).$$

For a reached target record s , with posterior $\pi = \pi_{os}^L$, it follows that

$$\sum_r K(o, r | a) T(s | o, r) = \frac{p_L(o, s)}{M(o, \pi)} \frac{\pi(a)}{q(a)} M(o, \pi) = L(o, s | a).$$

If s is unreached, full support makes both sides zero. Thus postprocessing K by T gives L . The symmetric construction gives a processor from L to K . Axiom (A4) and Lemma 1(a) give indifference. For a boundary prior, Lemma 1(b) first deletes the null actions.

For part (b), take the quotient of actual finite marked experiments by equality of (4). If E and G are two such experiments, their public mixture first draws a coin independent of the action, runs the selected experiment, and retains the coin in the record. Its marked law is

$$\eta_{tE \oplus (1-t)G} = t\eta_E + (1-t)\eta_G. \quad (6)$$

Axiom (A6) gives, for $0 < t < 1$ and any third experiment L ,

$$E \succeq_q L \iff tE \oplus (1-t)G \succeq_q tL \oplus (1-t)G. \quad (7)$$

For the reverse implication, if $E \not\succeq_q L$, completeness gives $L \succ_q E$; the strict part of Axiom (A6) makes the reverse public mixture strict. If record alphabets differ, put them in a common disjoint union before invoking the axiom; part (a) identifies the result with (6).

It remains only to check the Archimedean step. A closed segment between two experiments, and any fixed comparator, can be realised on one fixed union of their finite record alphabets. Its channel entries vary continuously with $t \in [0, 1]$. Axiom (A2) therefore makes both contour sets along the segment closed. Completeness says that they cover $[0, 1]$; when a target lies between the two endpoints, the sets contain opposite endpoints and connectedness makes them intersect. Thus every such target is indifferent to a point of the segment. The mixture-space theorem now supplies an affine representative W_q . This quotient is closed under every mixture and embedding used below. $\square$

A.2 The two common scales

Lemma 3 (Material utility and a common affine scale). *Under Axioms (A1)–(A7), there are outcomes o^+, o^- and a nonconstant function $u : O \rightarrow \mathbb{R}$ such that*

$$u(o^+) = 1, \quad u(o^-) = 0.$$

The payoff-lottery order is represented by $\sum_o \ell(o)u(o)$. The representatives in Lemma 2 may be normalised so that

$$W_q(\ell) = \sum_o \ell(o)u(o). \quad (8)$$

With this normalisation, adjoining a full-support independent dummy to a full-support prior preserves value, and for full-support q, q' ,

$$(q, E) \succeq (q', G) \iff W_q(E) \geq W_{q'}(G). \quad (9)$$

Proof. On a singleton action set, terminal-law sufficiency identifies an experiment with its payoff lottery. Public-mixture independence and continuity, proved as in Lemma 2, give an affine utility on $\Delta(O)$. Axiom (A7) supplies $o^+ \succ o^-$. Normalise their values to one and zero and let $u(o)$ be the value of the degenerate lottery at o . Affinity on the finite simplex gives expected utility.

An action report may ignore its input and draw an arbitrary target prior. Axiom (A5), applied in both directions, therefore makes this payoff lottery order independent of the prior and action alphabet. Normalise each W_q at the same two deterministic payoffs. The payoff-lottery embedding is affine and order-reflecting, so positive-affine uniqueness gives (8).

Dummy lifting between full-support fibres is likewise an affine order embedding. Positive-affine uniqueness initially permits

$$W_{q \otimes s}(E^\uparrow) = cW_q(E) + d, \quad c > 0.$$

The two payoff anchors have values zero and one on both sides; hence $c = 1$ and $d = 0$. To compare E at q with G at q' , lift each to the common product prior $q \otimes q'$, using the other prior as the dummy. Equation (3), the replacement part of Lemma 1, and the common product representative give (9). $\square$

We now use the pure-record theorem proved in Appendix D. Apply it to the constant-payoff lift

$$K_P(o, r \mid a) = \mathbf{1}_{\{o=o^-\}} P(r \mid a).$$

This lift commutes with block formation, record processing, action reporting, limits on each finite simplex, and compounding. Thus Axioms (A1) and (A8) give the pure weak order and its positive orientation; Axiom (A2) gives continuity; Axiom (A3) gives block coherence; Axioms (A4) and (A5) give the two data processing conditions, including every null-row completion; and Axiom (A6) gives branchwise continuation. These are precisely the hypotheses of the pure-record theorem. Its conclusion is that every within-channel and block-supported comparison between such lifts is ordered by $I_{q,P}(A; R)$.

Lemma 4 (The trace scale). *Under Axioms (A1)–(A8), there is one number $\lambda > 0$, independent of the prior and all finite alphabets, such that every full-support, nontrivial action fibre and every experiment E that pays o^- surely satisfy*

$$W_q(E) = \lambda I_q(E). \quad (10)$$

Proof. Fix a nontrivial full-support q . On the constant- o^- face, W_q is affine by Lemma 2. For any experiment E , the finite entropy identity

$$I_q(E) = H(q) - \sum_{(o,\pi) \in \text{supp } \eta_E} \eta_E(o, \pi) H(\pi). \quad (11)$$

shows that mutual information depends only on the marked terminal law and is affine under public mixtures. The pure-record theorem says that W_q and I_q represent the same nonconstant order on the constant-low face. Positive-affine uniqueness therefore gives

$$W_q(K_P) = \lambda_q I_{q,P} + b_q, \quad \lambda_q > 0.$$

The uninformative experiment has mutual information zero and, by (8), value $u(o^-) = 0$. Thus $b_q = 0$.

The coefficient does not depend on q . Let q and q' be nontrivial full-support priors. Lift full revelation at q to $q \otimes q'$ by adding an independent dummy coordinate. Normalised value and mutual information are both unchanged. Since the common information value is $H(q) > 0$, cancellation gives $\lambda_{q \otimes q'} = \lambda_q$. Lifting full revelation from the other coordinate gives $\lambda_{q \otimes q'} = \lambda_{q'}$. Hence all such coefficients agree. Equivalently, fix the uniform prior on a two-point dummy alphabet and call its coefficient λ . Axiom (A8) fixes the strict orientation, so $\lambda > 0$.

Finally, any experiment that pays o^- surely is a pure record experiment after the constant payoff coordinate is deleted; restoring that coordinate changes neither its marked terminal law nor its mutual information. This proves (10). $\square$

A.3 The branch calculation

Let q be full support on a nontrivial A and let $P : A \rightarrow \Delta(Y)$ be a first-stage record channel. Put

$$m_y := \sum_a q(a) P(y | a).$$

For a deterministic payoff profile $g : Y \rightarrow O$, let $C(P, g)$ denote the compound that first runs P and then pays $g(y)$ with no further record. Write g_y^o for the profile that pays o at y and o^- elsewhere.

Lemma 5 (The reached-branch payoff increment). *For every $y \in Y$ and $o \in O$,*

$$W_q(C(P, g_y^o)) - W_q(C(P, o^-)) = m_y u(o). \quad (12)$$

Proof. If $m_y = 0$, changing the payoff on branch y does not change the marked terminal law, so both sides are zero. Suppose $m_y > 0$, and put

$$r_y(a) := \frac{q(a) P(y | a)}{m_y}, \quad S_y := \text{supp } r_y.$$

Let $\bar{r}_y$ be the full-support restriction of r_y to S_y . For a marked continuation E on S_y , extend E to A through a support projection, insert it at branch y , and put the uninformative payoff o^- at every other branch. Denote the resulting experiment by $J_y(E)$. If $a \notin S_y$, then $r_y(a) = 0$; since $q(a) > 0$ and $m_y > 0$, the definition of r_y gives $P(y | a) = 0$. Hence the continuation rows chosen outside S_y are never used on branch y . Support restriction identifies their local versions as well, so the extension is independent of the chosen projection.

Support restriction and Axiom (A6), including its strict part, give

$$E \succeq_{\bar{r}_y} G \iff J_y(E) \succeq_q J_y(G).$$

For the reverse implication, failure of the local weak comparison gives a strict reverse comparison by completeness, and strict branchwise consistency gives a strict reverse compound comparison. Different continuation record alphabets are first put in a common disjoint union. Moreover, J_y commutes with public mixtures at the level of marked terminal laws. Thus affine uniqueness gives constants $c_y > 0, d_y$ such that

$$W_q(J_y(E)) = c_y W_{\bar{r}_y}(E) + d_y. \quad (13)$$

Suppose first that S_y has at least two elements. At payoff o^- , compare full revelation, E^{id} , with an uninformative record, E^0 . Lemma 4 gives the local difference

$$W_{\bar{r}_y}(E^{\text{id}}) - W_{\bar{r}_y}(E^0) = \lambda H(\bar{r}_y).$$

The finite mutual-information chain rule for insertion is

$$I_q(J_y(E)) = I_q(A; Y) + m_y I_{\bar{r}_y}(E). \quad (14)$$

Support extension does not change the last term. Both inserted experiments pay o^- surely, so Lemma 4 and (14) make their aggregate value difference $\lambda m_y H(\bar{r}_y)$. Equation (13) makes the same difference $c_y \lambda H(\bar{r}_y)$. Since $\lambda > 0$ and a nontrivial full-support distribution has positive entropy, $c_y = m_y$.

Locally, deterministic payoffs o and o^- have values $u(o)$ and zero by (8). Subtracting their two instances of (13), and identifying the resulting insertions with the two deterministic compounds in the statement, proves (12).

If S_y is a singleton, append an independent full-support two-point dummy action to the first-stage action. The reached posterior becomes the product of r_y and the dummy prior and hence has nontrivial support. The branch mass is still m_y , and Lemma 3 says that both compound values are unchanged. Apply the preceding argument in the product problem and remove the dummy. $\square$

Lemma 6 (Deterministic branch assembly). *For every deterministic payoff profile $g : Y \rightarrow O$,*

$$W_q(C(P, g)) = \lambda I_q(A; Y) + \sum_y m_y u(g(y)). \quad (15)$$

Proof. First observe that the increment from changing one branch is independent of the payoffs on the other branches. Indeed, for any two backgrounds g, h , the two crossed public half-mixtures

$$\frac{1}{2}C(P, g[y \mapsto o]) \oplus \frac{1}{2}C(P, h[y \mapsto o^-])$$

and

$$\frac{1}{2}C(P, g[y \mapsto o^-]) \oplus \frac{1}{2}C(P, h[y \mapsto o])$$

have the same marked terminal law. Away from y they contain the same two payoff-posterior atoms; at y they contain o and o^- with the same mass $m_y/2$. Affinity of W_q and cancellation of the two halves show that the two background-specific increments are equal. Taking h to be the

all-low profile and applying Lemma 5, then changing the finitely many branches one at a time, gives

$$W_q(C(P, g)) - W_q(C(P, o^-)) = \sum_y m_y u(g(y)). \quad (16)$$

The all-low compound pays o^- surely and has exactly the information in the first-stage record. Lemma 4 therefore gives $W_q(C(P, o^-)) = \lambda I_q(A; Y)$. Substitute this into (16). $\square$

A.4 Proof of the theorem

Proof of Theorem 1. Assume first Axioms (A1)–(A8). Let q have full support on a nontrivial action set, and let $K : A \rightarrow \Delta(O \times R)$ be arbitrary. Put $Y = O \times R$ and take as first stage

$$P_K(y | a) := K(y | a), \quad y = (o, r).$$

After $y = (o, r)$, pay o for sure and reveal nothing further. Call the resulting compound K^{seq} . From K , the record processor sends (o, r) to the sequential record $((o, r), *)$. In the reverse direction it sends $((o, r), *)$ to r ; inconsistent payoff-record pairs are unreached and may be completed arbitrarily. Both processors preserve the actual payoff coordinate. Hence Axiom (A4) gives $(q, K) \sim (q, K^{\text{seq}})$.

Apply Lemma 6 with $g(o, r) = o$. The two identities needed are

$$I_q(A; Y) = I_{qK}(A; O, R)$$

and

$$\sum_{o,r} p_{qK}(o, r) u(o) = \mathbb{E}_{q,K}[u(O)],$$

respectively. Together with sequentialisation, they give

$$W_q(K) = \mathbb{E}_{q,K}[u(O)] + \lambda I_{qK}(A; O, R). \quad (17)$$

If A is a singleton, mutual information is zero and the marked terminal law is that of the induced payoff lottery. Equation (8) gives (17) directly. If q is on the boundary, put $S = \text{supp } q$ and write (q^S, K^S) for the restriction. Then

$$(q, K) \sim (q^S, K^S), \quad V(q, K) = V(q^S, K^S). \quad (18)$$

The first identity is Lemma 1(b); the second follows directly from the finite sums defining expected utility and mutual information. Thus no representative W_q is assigned to a boundary fibre.

Restrict two arbitrary alternatives to their supports, apply (9) and (17) there, and use (18) to transport the comparison back. This gives

$$(q, K) \succeq (p, L) \iff V(q, K) \geq V(p, L), \quad (19)$$

with the same u and λ . Axiom (A3), first by duplication and then by deletion of irrelevant blocks, turns (19) into the within-channel representation and the finite-block moreover clause. The construction made u nonconstant and $\lambda > 0$. This proves the forward implication.

Conversely, suppose that a nonconstant u and a number $\lambda > 0$ represent the family by V . Axiom (A1) follows from the order on $\mathbb{R}$. Expected utility and mutual information are jointly continuous in (K, q) on each finite simplex; for the latter, use the entropy formula and $0 \log 0 = 0$. Hence the set of triples (K, q, p) for which $V(q, K) \geq V(p, K)$ is closed, which is Axiom (A2). A block-supported prior has a constant block label, which changes neither term of V ; this proves Axiom (A3) and reduces every pair comparison below to the corresponding two values.

A payoff-preserving record processor leaves the payoff marginal unchanged and weakly lowers $I(A; O, R)$ by data processing. This proves Axiom (A4). Under an action report, equation (2) again leaves the payoff-record marginal unchanged, and the joint law forms the Markov chain

$$(O, R) - A - B.$$

Thus $I(B; O, R) \leq I(A; O, R)$, proving Axiom (A5). This argument uses the undivided joint-law equation and so also covers every completion on a null reported action.

For a compound channel, the two finite chain rules are

$$\begin{aligned}\mathbb{E}[u(O)] &= \sum_{y:m(y)>0} m(y) \mathbb{E}_{q_y, K^y}[u(O)], \\ I(A; Y, O, R) &= I(A; Y) + \sum_{y:m(y)>0} m(y) I_{q_y, K^y}(A; O, R).\end{aligned}$$

The first-stage term is common to two continuation profiles. Their aggregate value difference is therefore

$$\sum_{y:m(y)>0} m(y) [V(q_y, K^y) - V(q_y, L^y)].$$

Branchwise weak improvement makes this sum nonnegative, and a strict improvement on a reached branch makes it positive. This is precisely Axiom (A6).

Because u is nonconstant, two deterministic payoffs are strictly ranked; the trace term is zero on a singleton action set. This gives Axiom (A7). At a constant payoff, full revelation has value $u(o) + \lambda H(q)$ and an uninformative record has value $u(o)$. Full support on a nontrivial finite action set implies $H(q) > 0$; hence Axiom (A8). The same block calculation used above gives the moreover clause with the original witnesses u and λ . $\square$